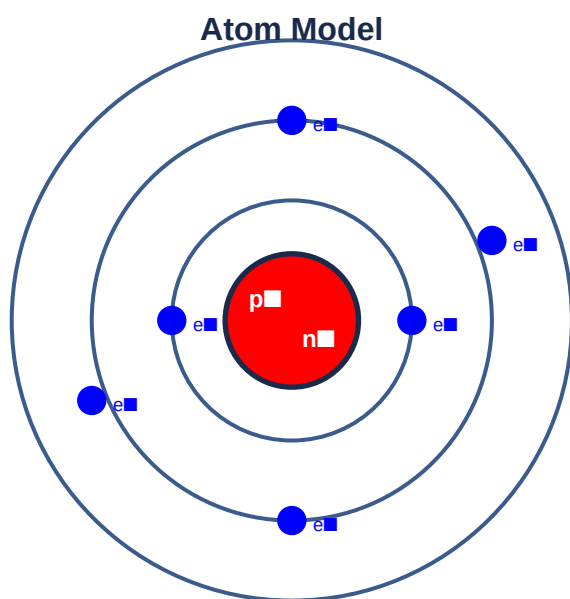


Unit 3: Structure of the Atom

Complete Study Notes – Grade 9 Chemistry



© Acadia. All rights reserved.

Introduction

The existence of atoms has been proposed since the time of early Indian and Greek philosophers (400 B.C.) who believed that atoms are the fundamental building blocks of matter. The word 'atom' comes from the Greek word 'a-tomos' which means 'uncuttable' or 'non-divisible'.

The atomic theory of matter was first proposed on a firm scientific basis by **John Dalton**, a British school teacher, in 1808. His theory, called **Dalton's Atomic Theory**, regarded the atom as the ultimate particle of matter.

3.1 Historical Development of Atomic Theories

Greek Philosophers and the Concept of 'Atomos'

In the middle of the 5th century BC, the Greek philosopher **Empedocles** proposed that all materials are made up of four **elements**: earth, air, water, and fire. **Plato** adopted this theory and coined the term 'element'. **Aristotle** also adopted the concept of four elements and introduced the idea that elements can be differentiated based on properties such as hot vs cold, and wet vs dry.

Democritus (460–370 BC) taught the theory of **atomism**, which held the belief that indivisible and indestructible atoms are the basic components of all matter in the universe. He believed that atoms are:

- **Completely solid** – no void inside an atom itself.
- **Homogeneous** – with no internal structure.
- **Different in size, shape, and weight** – explaining the variety of matter.

However, Aristotle and other prominent thinkers strongly opposed Democritus' theory, so it was disregarded and essentially buried until the 16th and 17th centuries.

Drawbacks of Early Greek Philosophers:

The early Greek philosophers tried to understand the world through reason and logic, but not through experiment and observation. They felt no need to justify their ideas based on life experiences. It took over two millennia before the theory of atoms was fully appreciated.

3.2 Fundamental Laws of Chemical Reactions

3.2.1 The Law of Conservation of Mass

The **Law of Conservation of Mass** (also known as the Law of Indestructibility of Matter) states that in any given system that is closed to the transfer of matter, the amount of matter in the system stays unchanged.

This law was discovered by **Antoine Lavoisier** in 1789. It states that **mass is neither created nor destroyed** in chemical reactions. In other words, the total mass of reactants equals the total mass of products.

Example: In the reaction: **Reactant 1 + Reactant 2 → Product 1 + Product 2**, the total mass of reactants equals the total mass of products.

Experiment demonstrating the Law: When silver nitrate solution (AgNO_3) reacts with dilute hydrochloric acid (HCl), a white precipitate of silver chloride (AgCl) is formed. The mass before

and after the reaction remains the same.

3.2.2 The Law of Definite Proportions (Proust's Law)

The **Law of Definite Proportions** states that every chemical compound contains fixed and constant proportions (by mass) of its constituent elements.

Example: Carbon dioxide (CO_2) is composed of one carbon atom and two oxygen atoms. The ratio of carbon to oxygen by mass is always 12:32, or simplified as 3:8.

Example: Water (H_2O) always contains hydrogen and oxygen in a mass ratio of 1:8, regardless of its source.

3.2.3 The Law of Multiple Proportions

The **Law of Multiple Proportions** states that when two different compounds are formed from the same two elements, the masses of one element that react with a fixed mass of the other are in a ratio of small whole numbers.

Example: Carbon monoxide (CO) has 1.33 g of oxygen combined with 1.00 g of carbon. Carbon dioxide (CO_2) has 2.66 g of oxygen combined with 1.00 g of carbon. The ratio of oxygen masses is $1.33:2.66 = 1:2$, a small whole number ratio.

3.3 Atomic Theory

3.3.1 Dalton's Atomic Theory

John Dalton (1803) proposed the first scientific atomic theory based on observations and experimental results.

The five postulates of Dalton's Atomic Theory:

1. **Elements are made of small particles called atoms.**
2. **Atoms can neither be created nor destroyed.**
3. **All atoms of the same element are identical** – they have the same mass and size.
4. **Atoms of different elements have different masses and size.**
5. **Atoms combine in small whole numbers to form compounds.**

Drawbacks of Dalton's Atomic Theory:

- **Atoms are divisible** – they can be further divided into protons, neutrons, and electrons.
- **Atoms of the same element can have different masses** – these are called **isotopes** (e.g., chlorine-35 and chlorine-37).
- **Atoms of different elements can have the same mass** – these are called **isobars** (e.g., argon-40 and calcium-40).
- **Atoms do not always combine in simple whole-number ratios** – complex organic compounds like sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) have larger ratios.

3.3.2 Modern Atomic Theory

The discoveries of subatomic particles led to the modified Modern Atomic Theory:

1. **Elements are made of small particles called atoms.**
2. **Atoms cannot be created or destroyed during ordinary chemical reactions.**

3. **All atoms of the same element have the same atomic number** (number of protons), but may vary in mass number due to isotopes.
4. **Atoms of different elements are different.**
5. **Atoms combine in small whole numbers to form compounds.**

3.4 Discoveries of Fundamental Subatomic Particles

3.4.1 Discovery of the Proton

The presence of positively charged particles in an atom was predicted by **Eugene Goldstein** (1886). He used a **cathode ray tube with a perforated cathode** and observed **anode rays** (canal rays) travelling in the opposite direction to cathode rays.

When the tube contained hydrogen gas, the particles of the canal rays were the lightest and had the highest charge-to-mass ratio. **Rutherford** showed these particles were identical to the hydrogen ion (hydrogen atom minus one electron). These particles were named **protons**.

Properties of anode rays:

- Travel in straight lines.
- Consist of material particles.
- Deflected in electric and magnetic fields (opposite to cathode rays).
- The nature (charge-to-mass ratio) depends on the gas in the tube.
- They are simply positively charged gaseous ions.

3.4.2 Discovery of the Electron

The electron was discovered by the British physicist **J.J. Thomson** in 1897 using a **Crooke's discharge tube** (cathode ray tube).

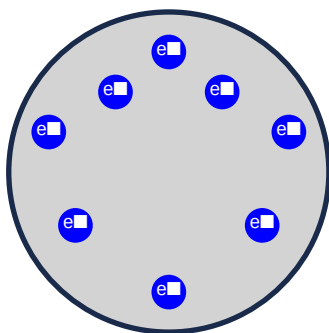
Thomson's Experiments on Cathode Rays:

1. **Straight-line travel:** A small object placed between cathode and anode created a sharp shadow on the tube, showing that cathode rays travel in straight lines.
2. **Particle nature:** A light paddle wheel placed between cathode and anode rotated when cathode rays fell on it, showing that cathode rays consist of particles with mass and kinetic energy.
3. **Negative charge:** When passed through an electric field, cathode rays bent towards the positive plate, proving they are negatively charged.
4. **Magnetic field:** When a magnetic field was applied, cathode rays deflected, further confirming they are negatively charged.

Thomson calculated the **charge-to-mass ratio** of the electron as 1.76×10^{-18} C/g. He concluded that atoms were not the most basic form of matter – they contain negatively charged particles called **electrons**.

Thomson's Plum Pudding Model

(Positive matrix with embedded electrons)



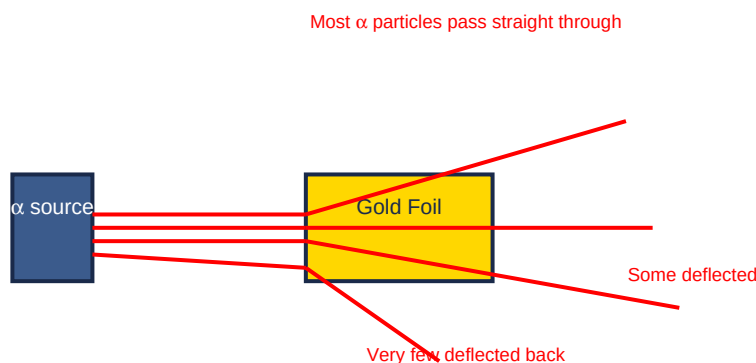
Thomson proposed the '**Plum Pudding Model**' of the atom in 1904. In this model, the electrons were embedded in a uniform sphere of positive charge, like blueberries stuck into a muffin. A watermelon is a good analogy – the seeds represent electrons, and the red flesh represents the positive charge.

Robert Millikan (1909) performed the **oil drop experiment** and determined the charge of the electron as 1.59×10^{-19} C. Using Thomson's charge-to-mass ratio, Millikan calculated the mass of the electron as 9.1×10^{-31} g.

3.4.3 Discovery of the Nucleus

Ernest Rutherford (1911) performed the famous **gold foil experiment** with his students Geiger and Marsden. They bombarded a very thin gold foil with α -particles (positively charged particles).

Rutherford's Gold Foil Experiment



Observations:

- **Most α -particles** passed straight through the foil – showing that most of the atom is empty space.
- **Some α -particles** were deflected at small angles.
- **A few α -particles** were deflected at large angles (over 90°).
- **Very few** (1 in 10,000) were reflected straight back.

Conclusions:

- The atom is mostly **empty space**.
- The **positive charge** and most of the **mass** of the atom are concentrated in a tiny central region called the **nucleus**.
- The diameter of the nucleus is about 10^{-13} cm, while the atom is about 10^{-8} cm – the nucleus is 10^5 times smaller than the atom.

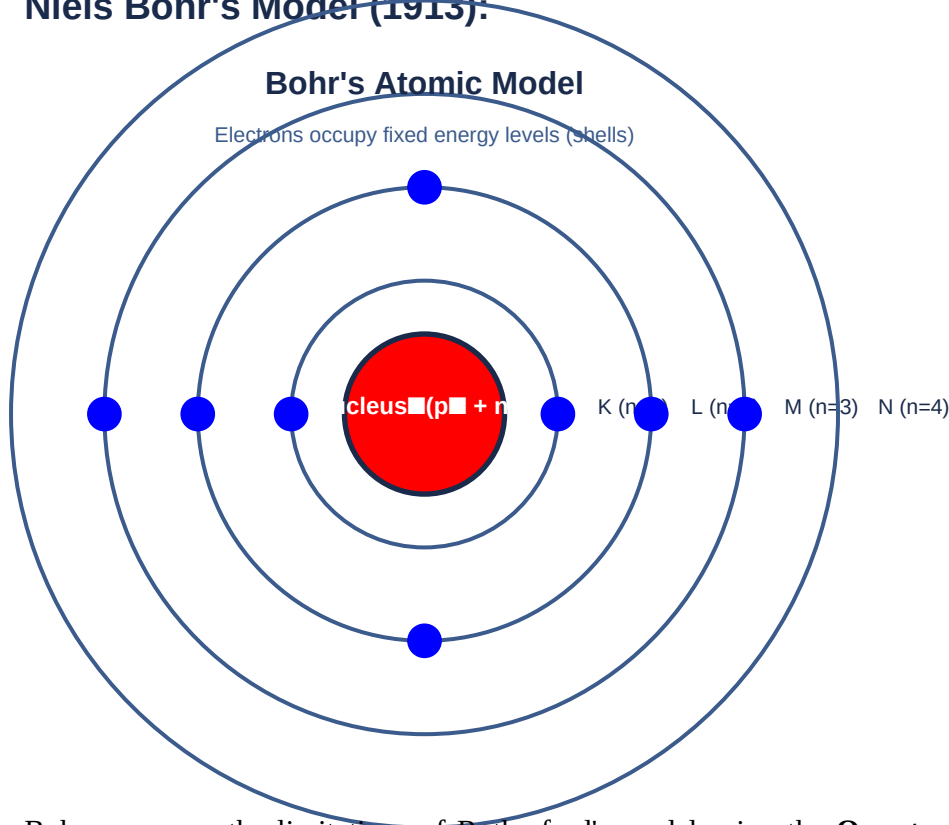
Rutherford's Atomic Model (Planetary Model):

- The atom consists of a small, positively charged **nucleus** at the centre.
- **Electrons** revolve around the nucleus in circular orbits, like planets around the Sun.
- The **electrostatic force** of attraction between the nucleus and electrons is balanced by the **centrifugal force** of the revolving electrons.

Drawbacks of Rutherford's Model:

- According to classical electrodynamics, a charged particle in circular motion **loses energy** and would spiral into the nucleus – the atom should collapse.
- It could not explain the **atomic spectra** (line spectra) of elements.

Niels Bohr's Model (1913):



Bohr overcame the limitations of Rutherford's model using the **Quantum Theory of Radiation** proposed by Max Planck. The key principles of Bohr's model are:

- Electrons are arranged in **concentric circular orbits** around the nucleus.
- Electrons occupy only **certain orbits** called **stationary orbits** or **shells** (K, L, M, N).
- Each orbit is associated with a **fixed energy** – these are called **energy levels**.
- The energy increases as the distance from the nucleus increases: $K < L < M < N$.
- As long as an electron revolves in a particular orbit, it does **not lose or gain energy**.

3.4.4 Discovery of the Neutron

The neutron was discovered by **James Chadwick** in 1932. He bombarded beryllium nuclei with α -particles and observed radiation that:

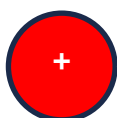
- Could rotate a paddle wheel – showing it had material particles.
- Was **not deflected** by an electric or magnetic field – showing it had **no charge**.

These neutral particles were called **neutrons**. They have a mass slightly greater than that of a proton, and they reside in the nucleus along with protons.

3.5 Composition of an Atom and Isotopes

Subatomic Particles

Proton (p) $\blacksquare$ +1 charge $\blacksquare$ 1.0073 amu Neutron (n) $\blacksquare$ 0 charge $\blacksquare$ 1.0087 amu Electron (e) $\blacksquare$ -1 charge $\blacksquare$ 0.00055 amu



Particle	Symbol	Charge	Mass (amu)	Mass (kg)	Location
Proton	p $\blacksquare$	+1	1.0073	1.673×10^{-27} $\blacksquare$	Inside nucleus
Neutron	n $\blacksquare$	0	1.0087	1.675×10^{-27} $\blacksquare$	Inside nucleus
Electron	e $\blacksquare$	-1	0.00055	9.1×10^{-31} $\blacksquare$	Outside nucleus

3.5.1 Electrons, Protons and Neutrons

Electrons: Extremely small particles (0.00055 amu) with a -1 charge. They contribute almost nothing to the total mass of an atom.

Protons: Positively charged (+1) particles with a mass of 1.0073 amu. They are found in the nucleus.

Neutrons: Neutral particles (no charge) with a mass of 1.0087 amu. They are found in the nucleus alongside protons.

3.5.2 Atomic Number and Mass Number

Atomic number (Z) = the number of protons in the nucleus of an atom.

Mass number (A) = the total number of protons and neutrons in the nucleus.

$$A = Z + N \text{ where } N = \text{number of neutrons.}$$

For a neutral atom: **Number of protons** = **Number of electrons**.

Nuclear Symbol:

The mass number is written as a superscript (upper left) and the atomic number as a subscript (lower left) of the chemical symbol.

Example: ^{12}C – carbon-12, with 6 protons and 6 neutrons.

Example: How many protons, electrons, and neutrons are in an atom of potassium (K) with a mass number of 40?

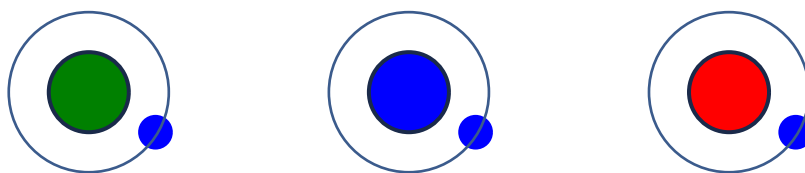
Solution: Atomic number of K = 19 $\rightarrow$ 19 protons. Neutral atom $\rightarrow$ 19 electrons. Neutrons = Mass number – Atomic number = 40 – 19 = 21 neutrons.

3.5.3 Atomic Mass and Isotopes

Isotopes are atoms of the same element that have the same number of protons but different numbers of neutrons. They have the same atomic number but different mass numbers.

Isotopes of Hydrogen

Protium ^1H 1 proton 0 neutrons Deuterium ^2H 1 proton 1 neutron Tritium ^3H 1 proton 2 neutrons



Example: Hydrogen Isotopes

- **Protium (^1H)** – 1 proton, 0 neutrons, mass number 1.
- **Deuterium (^2H)** – 1 proton, 1 neutron, mass number 2.
- **Tritium (^3H)** – 1 proton, 2 neutrons, mass number 3 (radioactive).

Atomic Mass Unit (amu):

One atom of carbon-12 is assigned a mass of **12 atomic mass units (amu)**. $1 \text{ amu} = 1.6605 \times 10^{-24} \text{ g}$.

Average Atomic Mass:

The average atomic mass of an element is a **weighted average** of the masses of its isotopes, based on their natural abundances.

$$\text{Average Atomic Mass} = [\% \text{ isotope 1} \times \text{mass isotope 1}] + [\% \text{ isotope 2} \times \text{mass isotope 2}] + \dots$$

Example: Lithium has two naturally occurring isotopes: Li-7 (93%) and Li-6 (7%). Average atomic mass = $(7 \times 0.93) + (6 \times 0.07) = 6.51 + 0.42 = 6.93 \text{ amu}$.

Example: Boron has B-10 (20%) and B-11 (80%). Average atomic mass = $(10 \times 0.20) + (11 \times 0.80) = 2 + 8.8 = 10.8 \text{ amu}$.

3.5.4 Main Energy Levels

The **principal energy level** (main energy level) of an electron refers to the shell in which the electron is located relative to the atom's nucleus. These are denoted by the **principal quantum number** (n).

Shells are designated as K, L, M, N, ... or 1, 2, 3, 4, ...

Shell	Principal Quantum Number (n)	Maximum Electrons ($2n^2$)
K	1	2
L	2	8
M	3	18
N	4	32

3.5.5 Electronic Configuration on Main Shells

The **electronic configuration** of an element describes how electrons are distributed in their energy levels (shells). Electrons fill the shells starting from the lowest energy shell (K) and moving outward.

Rules for electronic configuration:

- The maximum number of electrons in a shell is $2n^2$.
- The **valence shell** (outermost shell) can hold a maximum of 8 electrons.
- Electrons fill shells from the lowest energy (K, L, M, N, ...) upward.
- The **penultimate shell** (second from outermost) can hold up to 18 electrons.
- The **antipenultimate shell** (third from outermost) can hold up to 32 electrons.

Examples:

- Hydrogen (H): $1 \rightarrow 1$ electron $\rightarrow 1$
- Helium (He): $2 \rightarrow 2$ electrons $\rightarrow 2$
- Lithium (Li): $3 \rightarrow 2, 1$
- Carbon (C): $6 \rightarrow 2, 4$
- Neon (Ne): $10 \rightarrow 2, 8$
- Sodium (Na): $11 \rightarrow 2, 8, 1$
- Calcium (Ca): $20 \rightarrow 2, 8, 8, 2$

3.5.6 Valence Electrons

Valence electrons are the electrons that occupy the outermost shell of an atom. They determine the element's chemical properties and how an atom will react during chemical reactions.

Example: Silicon (Si) has electronic configuration 2, 8, 4. The 4 electrons in the M shell are **valence electrons**.

Example: Argon (Ar) has electronic configuration 2, 8, 8. The 8 electrons in the M shell are **valence electrons** – this makes argon very stable (inert gas).

Unit Summary

- The word 'atom' comes from the Greek 'atomos' meaning 'uncuttable'.
- **Dalton's Atomic Theory** (1803) proposed that atoms are indivisible and indestructible.
- **Fundamental laws:** Law of Conservation of Mass (Lavoisier), Law of Definite Proportions (Proust), Law of Multiple Proportions (Dalton).
- **Proton** – discovered by Goldstein (1886), positive charge (+1), mass 1.0073 amu, located in nucleus.

- **Electron** – discovered by J.J. Thomson (1897), negative charge (-1), mass 0.00055 amu, located outside nucleus.
- **Neutron** – discovered by James Chadwick (1932), neutral charge, mass 1.0087 amu, located in nucleus.
- **Thomson's model** (Plum Pudding): electrons embedded in a positive sphere.
- **Rutherford's model** (Planetary): atom has a small, dense, positively charged nucleus with electrons orbiting around it.
- **Bohr's model**: electrons occupy fixed energy levels (shells) – K, L, M, N.
- **Atomic number** = number of protons. **Mass number** = number of protons + neutrons.
- **Isotopes** – same element, same atomic number, different mass numbers (different number of neutrons).
- **Average atomic mass** – weighted average of the masses of all isotopes.
- **Electronic configuration** – distribution of electrons in shells, starting from K shell.
- **Valence electrons** – electrons in the outermost shell, determine chemical properties.

Key Terms

Atomic mass – the average mass of an atom of an element, measured in amu.

Atomic mass unit (amu) – 1/12 the mass of a carbon-12 atom.

Atomic number (Z) – the number of protons in the nucleus.

Atomic theory – the theory that matter is composed of atoms.

Cathode ray – a stream of electrons produced in a discharge tube.

Electron – a negatively charged subatomic particle.

Electronic configuration – the distribution of electrons in energy levels.

Isotope – atoms of the same element with different numbers of neutrons.

Mass number (A) – the total number of protons and neutrons in the nucleus.

Neutron – a neutral subatomic particle found in the nucleus.

Nucleus – the dense, positively charged centre of an atom.

Proton – a positively charged subatomic particle found in the nucleus.

Valence electrons – electrons in the outermost shell of an atom.

End of Unit 3 – Structure of the Atom